

DIRECTORY

The flourishing relationship between science and industry during the 19th century fueled the rise of science as a profession. Backed by strong academic institutions, scientists shared information and were able to begin to unify individual discoveries into coherent wider theories.

ALEXANDER VON HUMBOLDT 1769–1859

Born in Berlin, Prussia (modern-day Germany), Alexander von Humboldt was an originator of modern geography and ecological thinking. He was one of the first scientists to study how physical conditions—such as climate, altitude, latitude, and soils—affect the distribution of life. A great explorer as well as a scientist, he spent 5 years in South America, where he helped map how flora and fauna differed between sea level and high altitude in the Andes. He also noted that the rotation of Earth did not affect the direction of ocean currents, and he proposed that Africa and South America were once linked. After returning to Europe, he spent much of his life writing up his findings in a five-volume work called *Kosmos*.

HANS CHRISTIAN ØRSTED 1777–1851

Danish scientist Hans Christian Ørsted made his pivotal discovery linking magnetism and electricity while lecturing at the University of Copenhagen. He was showing how the electric current from a voltaic pile (an early battery) can heat up a wire and make it glow when he

noticed that a compass needle standing near the wire moved every time the current was switched on. Further study convinced him that the current produced a circular magnetic field as it flowed through the wire. Ørsted's discovery changed how scientists understood electricity and magnetism, which were previously considered as two separate forces of nature. Further developed by Michael Faraday and James Clerk Maxwell, it led to the modern theory of electromagnetism.

HUMPHRY DAVY 1778–1829

After a brief career studying gases in Bristol's Pneumatic Institution, English chemist Humphry Davy was invited to join the Royal Institution in London. It was here he discovered new elements, including potassium and sodium, through a process that later became known as electrolysis. Davy realized that the electricity produced by a voltaic pile (an early battery) is the result of the chemical reactions happening inside it and that reversing the process would allow electricity to split chemical compounds into the elements from which they were made. Many previously unknown elements were discovered in this way.

FRIEDRICH WOHLER 1800–1882

Born near Frankfurt in what is now Germany, Friedrich Wöhler graduated in medicine but became a distinguished chemist after studying under Jöns Jacob Berzelius in Sweden. His reputation grew first through his purification of aluminum. Then, in 1828, he accidentally created urea from ammonia and cyanic acid. Previously, it was believed that biological, organic compounds such as urea could only be obtained from a living organism.

JUSTUS VON LIEBIG 1803–1873

German chemist Justus von Liebig made many advances in the chemistry of organic substances, which at the time meant substances made in living organisms. Liebig maintained that the secret of organic chemistry lay in knowing that the properties of substances containing just carbon and hydrogen (and sometimes oxygen and nitrogen as well) were as variable as all other elements put together. He later asserted that chemists would one day be able to synthesize sugar, as well as the painkillers salicin and morphine.

RICHARD OWEN 1804–1892

The first person to coin the word “dinosaur,” Richard Owen used the term *Dinosauria* (meaning “terrible lizard”) to